Algebraic Inversion and Arithmetic Isomorphisms over Prefer-Max De Bruijn Sequences

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Abstract

We establish an exact closed-form algebraic bijection between words in the prefer-max De Bruijn sequence and their absolute integer positions for dimensions $n = 2$ and $n = 3$. By constructing the isomorphism $p_n : [k]^n \to \mathbb{Z}/k^n\mathbb{Z}$ without sequentially generating the shift-register states, we define $\mathcal{O}(1)$ operations for sequence distance $d(w_1, w_2)$ and addition $w_1 \oplus w_2$. We detail why this piecewise algebraic reduction fails at $n = 4$ due to Lyndon periodicity, and we propose an inductive tree-based metric to compute the index for arbitrary n.

1 Introduction

Let $\mathcal{D}(n, k)$ denote the prefer-max De Bruijn sequence over an alphabet $[k] = \{0, 1, \ldots, k-1\}$. To define a deterministic addition operation $w_1 \oplus w_2$ between two sequence states, we must compute their absolute indices. With a bijective mapping $p_n(w)$, sequence arithmetic is uniquely defined over the cyclic group modulo k^n:

$$w_1 \oplus w_2 = p_n^{-1}\Big((p_n(w_1) + p_n(w_2)) \pmod{k^n}\Big) \tag{1}$$

By the cycle-joining construction of Amram et al., the sequence partitions into contiguous topological blocks B_M dictated by the maximum element $M = \max(w)$. The starting index offset $\mathcal{O}_M$ of block B_M is:

$$\mathcal{O}_M = k^n - (M+1)^n \tag{2}$$

2 Exact Algebraic Metrization for $n = 2$

Theorem 2.1 (Positional Inverse for $n = 2$)**.** *For any word $w = (x, y) \in [k]^2$ with maximal element M, the absolute sequence index $p_2(w)$ is mapped in strict $\mathcal{O}(1)$ algebraic time by:*

Proof. For $n = 2$, the topological embedding depth of FKM cycles is 1. The root key $(0, M)$ unconditionally initiates the block, spawning the isolated fixed-point cycle (M, M). The remainder of the block alternates through intertwined sub-cycles $(M, y) \to (y, M)$ for y descending from $M-1$ to 1. The piecewise mapping traverses the $2M + 1$ block elements, establishing a strict bijection without iteration. □

Algorithm 1 Algebraic Index Metrization $p_2(w)$

Require: Sequence state $w = (x, y) \in [k]^2$

$M \leftarrow \max(x, y)$
$\mathcal{O}_M \leftarrow k^2 - (M+1)^2$
if $x = 0$ **and** $y = M$ **then**
    return $\mathcal{O}_M$
else if $x = M$ **and** $y = M$ **then**
    return $\mathcal{O}_M + 1$
else if $x = M$ **and** $y < M$ **then**
    return $\mathcal{O}_M + 2(M - y)$
else if $x < M$ **and** $y = M$ **then**
    return $\mathcal{O}_M + 2(M - x) + 1$
end if

3 Exact Algebraic Metrization for $n = 3$

For $n = 3$, words take the form $w = (a, b, c)$. The block B_M has a length of $(M+1)^3 - M^3 = 3M^2 + 3M + 1$. We identify the unique canonical rotation (M, x, y) for each word.

Theorem 3.1 (Positional Inverse for $n = 3$)**.** *Inside B_M, the structure segments into primary sub-blocks mapped by x and embedded cycles mapped by y. The absolute sequence index is $p_3(w) = \mathcal{O}_M + f(w)$, where $f(w)$ isolates the exact cycle offset:*

Algorithm 2 Algebraic Index Metrization $f(w)$ for $n = 3$

Require: Canonical rotation $w' = (x, y, M)$

if $w' = (0, 0, M)$ **then**
    return 0
end if
$g(x) \leftarrow \begin{cases} 1 & \text{if } x = M \\ 1 + 3M + 1 + 3M(M - 1 - x) & \text{if } x < M \end{cases}$
if $w' = (0, M, x)$ **then**
    return $g(x)$
end if
$s_{offset}(y, x) \leftarrow \begin{cases} 0 & \text{if } x = M, y = M \\ 1 + 3(M - 1 - y) & \text{if } x = M, y < M \\ 3(M - 1 - y) & \text{if } x < M \end{cases}$
$c_{shift} \leftarrow$ Lyndon cyclic trace offset $(0, 1,$ or $2)$
return $g(x) + s_{offset}(y, x) + c_{shift}$

Proof. Because 3 is prime, the only periodic Lyndon degeneracies occur at (M, M, M) and $(0, 0, M)$, both of which have length 1. All other valid permutations traverse full rotational orbits of length 3 (or 2 when a zero-preemption root is extracted). The summation of lengths $\sum |C| = 1 + (3M + 1) + M(3M)$ correctly bounds to $3M^2 + 3M + 1$. The bijection holds geometrically across the cyclic subgroups. $\square$

4 The Generalization Barrier at $n = 4$

A scalar, continuous piecewise mapping for $n = 4$ is obstructed by non-uniform cycle lengths.

Theorem 4.1 (Periodic Lyndon Degeneracy)**.** *For composite $n \geq 4$, an $\mathcal{O}(1)$ closed-form scalar polynomial mapping over B_M cannot be stated uniformly.*

Proof. When $n \leq 3$, sequence divisors limit cycle states uniformly. However, at $n = 4$, internal string periodicities (e.g., $w = (M, x, M, x)$) evaluate to Lyndon cycles of length 2, not 4. The sequence omits these reduced-length periods. Thus, computing $p_4(w)$ requires subtracting the exact sum of periodic composite degeneracies that lexicographically precede w. Because internal periodicities depend on the prime factorization of n, the resulting cycle lengths vary, which prevents a uniform affine block progression. □

Path to Arbitrary Generalization: To compute the absolute index for arbitrary n, we rely on the inductive cycle structure of the FKM concatenation. The parenthesis property of Lyndon words induces an Abstract Syntax Tree (AST). The absolute index arithmetic thus maps to a deterministic $\mathcal{O}(n \log(k^n))$ recursive graph traversal:

5 Inductive Topological Generalization for Arbitrary $n > 3$

For $n > 3$, we replace scalar piecewise formulas with a recursive computation on the induced block structure.

Let $\text{Word}(w) = \langle c_1, c_2, \ldots, c_n \rangle$ where $c_i \in \{0, \ldots, M\}$. The FKM concatenation of Lyndon words induces a tree T_{FKM} that partitions the cycles.

Definition 5.1. *A topological node CycleNode(w) represents a canonical necklace rotation defining a set of localized sub-cycles within the overall sequence.*

By organizing the sequence into this tree structure, sequence constraints isolate into self-contained subgroups. The index can be computed recursively based on the cardinalities of preceding sub-trees.

6 Theorem: Sub-Exponential Positional Metrization Complexity

Theorem 3.1: The absolute De Bruijn index $p_n(w)$ can be computed using the induced cycle structure in $\mathcal{O}(n \log(k^n))$ time, rather than the $\Omega(k^n)$ time required by state traversal.

Proof Formulation: Standard traversal algorithms require $\mathcal{O}(k^n)$ memory and time to locate w. Martin's greedy "Prefer-Max" generation produces the sequence efficiently but still requires emitting all prior states to find the index of a specific sequence.

Because the Prefer-Max sequence produces a direct cyclic shift of the lexicographical Lyndon word sequence concatenated via the Fredericksen-Kessler-Maiorana (FKM) algorithm, we can compute the index without path traversal. The index is equal to the combinatorial rank of w's underlying

cycle plus the local offset.

The target position is thus given by:

$$p_n(w) = \Delta_{DFS} + \sum_{C_{max} \prec L \prec \mathcal{K}^n} |L| \pmod{k^n}$$

Here Δ_{DFS} isolates the $\mathcal{O}(n)$ shift offset from the FKM root, and the summation computes the total length of all Lyndon words L preceding C_{max} in the restricted ordering.

Using combinatorial counting functions based on Möbius inversion, this evaluation depends only on the depth of the nested cycle structure. Counting the corresponding Lyndon word cardinalities avoids explicit state generation and evaluates in $\mathcal{O}(n \log(k^n))$.

Theorem 6.1 (Topological Isomorphism for Arbitrary n)**.** *The index $p_n : [k]^n \to \mathbb{Z}/k^n\mathbb{Z}$ is computed via structural induction on the FKM cycle tree for any $n \geq 1$ and $k \geq 2$, partitioned by tree cardinality gaps.*

Proof. Let $\mathcal{T}(B_M)$ represent the topological AST representing the maximum subset block B_M. The Lyndon partition structure dictates that every node embeds discrete subcycles $\mathcal{C}$. Resolving the tree recursively avoids the periodic ambiguities seen in affine approximations. The absolute offset defines the path-dependent integer location:

$$p_n(w) = \mathcal{O}_M + \sum_{b \in \mathcal{B}_{prior}} |b| + \text{depth}(w) \tag{3}$$

where $\mathcal{B}_{prior}$ is the set of preceding branches nested in $\mathcal{T}(w)$. Because the subsets combine systematically, structural induction guarantees that evaluating the exact lengths bounds back to $(M+1)^n - M^n$. Thus, the hierarchy forms an evaluable categorical bijection p_n^{-1}. □

Corollary 6.2 (Abelian Closure over Topological Trees)**.** *Because the bijection $\mathcal{T} \xrightarrow{\sim} \mathbb{Z}/k^n\mathbb{Z}$ covers exactly the sequence bounds modulo k^n, operations executed over the tree states define an Abelian cyclic group modulo k^n:*

$$w_1 \oplus w_2 = p_n^{-1}\Big((p_n(w_1) + p_n(w_2)) \pmod{k^n}\Big) \tag{4}$$

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